Effects of Performing Augmented Reality Tasks on Postural Sway While on a Ladder

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ABSTRACT

The use of augmented reality (AR) is growing in industrial occupational settings such as within the construction industry [1, 2]. Falls off ladders are already a major source of workplace injuries [3], and any adverse effects of AR on balance while working on a ladder could exacerbate the risk of falling. Therefore, the purpose of this study was to investigate the effects of viewing and interacting with AR on balance during simulated construction tasks while on a ladder. The four tasks investigated involved: (1) a reading task with AR input, (2) an assembly task requiring AR input, (3), an assembly task with verbal input, and (4) a button poking task with virtual AR buttons. We also investigated screen-relative and world-relative AR presentation styles across the relevant AR-support tasks. Balance was assessed by measuring mean speed of postural sway of the center of pressure with increases in speed suggesting a greater risk of falling.

Regarding the tasks, results indicated postural sway was greater during the assembly task with verbal input (6.39 cm/s; $p<0.05$), the assembly task with AR input (6.33 cm/s; $p<0.05$), and the button poking task with AR input (7.35 cm/s; $p<0.05$) as compared to the reading task with AR input (2.90 cm/s). Additionally, postural sway increased more during the button poking task with AR input as compared to the assembly task with AR input. Regarding the two presentation styles, postural sway with a world-relative AR display (9.27 cm/s) was greater than a screen-relative AR display (4.95 cm/s) during the poking task with AR input ($p < 0.05$), but did not differ between presentation styles during the other tasks ($p > 0.05$). Our results suggest that care must be taken when designing AR applications for applications that are to be used on ladders or other environments where postural stability is of great importance, as the task to be completed and the presentation style can affect a worker's balance. Further research into user interface presentations styles interaction techniques and tasks involving AR menu navigation is warranted.

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GENERAL AUDIENCE ABSTRACT

Falls from ladders cause tens of thousands of workplace injuries and hundreds of deaths across the United States every year. Workers in the construction industry in particular are beginning to use augmented reality (AR)—digital information overlaid onto the real world—to assist with tasks like reading instructions and hard-to-reach or overhead fine motor tasks. While AR has the potential to make work easier, little is known about how using an AR headset affects a worker's balance while standing on a ladder. Twenty-eight participants were asked to climb a ladder and complete four tasks while wearing a Microsoft HoloLens 2 AR headset. The tasks included reading text, poking virtual buttons, assembling colored blocks with AR visual guidance, and assembling blocks guided by oral instructions. A pressure plate beneath the ladder tracked how much participants' center of pressure changed during each task. Participants were given the opportunity to practice each task before completing the experimental trials.

Poking virtual buttons caused the greatest postural sway, likely because it required participants to reach forward into space, whereas reading AR text caused the least. How the AR content was displayed also mattered; a display fixed in space caused greater postural sway than one that stayed centered in the user's field of view, especially during the button poking tasks. These findings highlight design implications for how AR applications should be used in the construction industry. Interactive AR interfaces should favor displays that move with the user's head, and tasks requiring physical reaching should be approached with caution while on a ladder. This research opens discussion into safer AR application design for completing construction work on a ladder.

Table of Contents

1. INTRODUCTION

Augmented Reality (AR) is a spatial computing technology that uses projections to overlay virtual objects into physical environments [1, 2]. Within the construction industry, AR has found uses in installation, maintenance, repair, and inspection [3-5]. During these activities it is common to refer to paper instructions, manuals, and plans, which require a worker to hold in-hand or place nearby to access the information. AR enables workers to free both hands and engage with information presented in their environment [5, 6]. Research recommends use of AR for construction tasks because studies have shown that AR can reduce the cognitive load typically associated with remembering or spatially translating time-sensitive information and instructions [6].

Within construction and related occupations, it is common to use ladders to reach workspaces that are overhead or higher than a worker can comfortably reach. Injuries resulting from falls off ladders are common in the workplace, making up over 22,000 workplace injuries and 161 workplace fatalities in 2020 [7]. Within the construction industry alone, over 80% of injuries that required treatment in an emergency room involved falling off a ladder [8]. Therefore, it is reasonable to raise concerns over the possible safety effects of using AR devices while on a ladder. Given that virtual reality tools can lead to balance impairment due to a range of factors including the vergence-accommodation conflict, a similar effect may occur with AR [9-11]. As such, the use of AR while working on ladders may further increase fall risk.

In this work, we aim to determine to what extent usage of AR while working on a ladder affects postural sway, which is an indicator of balance and fall risk. Within the construction industry ladders are essential tools for completing work out of reach of the average worker. Balance impairment has also been reported in optical see through headsets [9, 12]. With an increased presence of AR tools in the workplace, it is essential to understand the safety implications of these devices.

The study presented herein examines four different tasks that aim to simulate potential AR use within the construction industry: a text reading task with AR input, a block assembly task with AR input, the same block assembly task with verbal input, and a button poking task with AR input. The three tasks relying on visual AR input are studied across two different presentation styles: screen-relative and world-relative. A screen-relative display remains centered in a user's field of vision while a world-relative display remains in a fixed location within the environment. Balance was assessed by measuring mean speed of postural sway of the center of pressure using a force plate under the ladder, with increases in sway speed from a baseline condition involving standing as still as possible on the ladder being compared between tasks and presentation styles.

The purpose of this study was to investigate the effects of viewing and interacting with AR on balance during simulated construction tasks while on a ladder. To the best of our knowledge, there have been no empirical investigations into a user's postural sway while using AR to work on a ladder. One specific aim is proposed:

Specific Aim: Evaluate differences between four AR-related tasks within two application display conditions on postural sway while working on a ladder. Participants will be asked to climb a ladder at a height of approximately four feet off the ground and perform four tasks while on the ladder: a projected reading task, a block-sorting task with projected visual instructions, a projected interactive button-poking task, and a block-sorting task with oral instructions. Apart from the block-sorting task with oral instructions, all tasks will have two different conditions: a world-relative display and a screen-relative display. A force plate beneath the ladder will record their center of pressure over time and provide an average value over a 30 second period. These values will be used to determine overall stability while performing the tasks. Two primary research questions are proposed:

Research Question 1: What is the effect of simulated AR work tasks on postural sway while working on a ladder?

Research Question 2: Do the effects of simulated AR work tasks on postural sway differ between screen-relative and world-relative presentation styles?

Two hypotheses will be tested:

Hypothesis 1: Compared to other AR tasks, the increase in postural sway from baseline will be the largest during the interactive button-poking task. This hypothesis was based on postural sway being dependent on mental facilities that are compromised when cognitive demand is increased, as dictated by multiple resources theory [12-14].

Hypothesis 2: The increase in postural sway from baseline during AR tasks will be larger with a screen-relative presentation style than with a world-relative presentations style. This hypothesis is based on postural sway being dependent on the visual, vestibular, and proprioceptive sensory systems [9, 15, 16]. Having a display constantly in the center of a participant's field of vision is expected to obscure the environment and cause participants to fixate on a target not grounded in the physical environment [15, 17].

While no studies have compared tasks and presentation styles on postural sway while working on a ladder, similar fields have been explored. By filling the current gap in understanding, this study will open opportunities for follow-up research into balance interventions, tailoring augmented reality tasks to allow for more stable conditions, and the types of tasks that can safely be performed on a ladder.

2. LITERATURE REVIEW

2.1. Augmented Reality

Historically the primary method of accessing digital information has been a desktop computer with a command-line interface [18]. This desktop computer stood alone, useless without the proper accessories to communicate with the machine—a monitor and a keyboard. These methods of human-computer interaction have been a cornerstone of how people interface with computers for decades; the monitor allows users to view the computer's state and read information, and the keyboard allows for a user to send commands and record information. After the advent of the graphical user interface around the 1980s, the need for a method of planar navigation gave us the computer mouse, and for many years, our methods of working with computers remained unchallenged.

While desktop computers are still prominent in the workplace and for private use, the demand for a computer that could be transported spurred the mobile computing domain, ushering in the laptop computer. This combined the computer, monitor, and keyboard, but the design of the traditional mouse had to change to allow every component to be combined into one device. As such, the trackpad created a flat surface that could interpret the motions of a finger as motion of a cursor on a monitor. As the desire to further shrink computers increased, the tablet computer and the smart phone took the world by storm in the 2000s, making computers even smaller [18]. The next accessory to be removed was the keyboard, opting for a visual keyboard that could be tapped instead of using physical switches for keys. After the dominance of the smart phone as the pinnacle of computing, the 2010s saw the mainstream rise of virtual, augmented, and mixed reality systems, and the paradigm of user interfaces once again shifted.

AR is a form of multimodal technology that overlays digital information and graphics overtop a physical environment in real time [1, 2]. This is achieved by one of three methods of projecting a digital environment atop the physical environment: video see-through, optical see-through, and projective displays [2]. These displays are often accompanied by auditory methods of feedback, whether by small speakers or headphones, aiding in their usage [13]. AR interfaces allow for a new paradigm of work that does not require traditional monitors, keyboards, and mice [18].

Technologies like AR differ from traditional desktop computing in a domain referred to as spatial computing. Spatial computing describes how overlays can now exist and be manipulated in reference to real-world objects and environments [18]. Computer vision (through cameras within headsets or within handheld devices) maps the user's surroundings and determines where virtual objects can exist, even if they are obscured or entirely hidden by physical objects in the real world, a phenomenon known as occlusion [18].

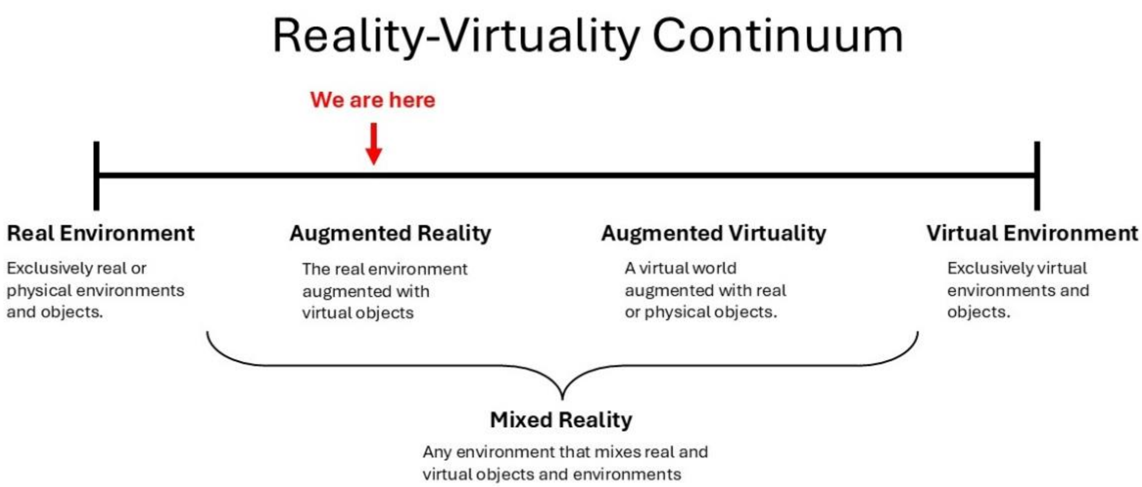

Figure 1. Reality-Virtuality Continuum. Adapted from A. Heinzel, S. Azhar, and A. Nadeem, "Uses of augmented reality technology during construction phase." [3].

A form of spatial computing that has garnered more attention in commercial markets is virtual reality (VR) headsets. AR differs from VR in a few important ways, one of which is the level of integration with the physical environment. Researchers who use AR and VR often refer to the Reality-Virtuality Continuum (Figure 1) to determine a technology's level of integration with the real environment [3]. While VR exists at the far right end of the continuum, AR exists closer to the left end, as the real environment is still visible and involved, but virtual objects are overlayed over the real environment [1, 19]. This allows for virtual objects to appear as though they are coexisting in real time within the physical environment [2].

2.2. Postural Sway and Sensory Systems

Balance, while a fundamental function of the body, is a complex process involving mechanical, physiological, anatomical, and psychological components [9]. The visual, vestibular, and proprioceptive sensory systems work together to help maintain upright balance and regain balance if perturbed. Any changes to the effectiveness of the visual, vestibular, and proprioceptive sensory systems (such as when vision is obscured with AR use), typically create greater postural sway due to an increase in reliance on cortical power from the prefrontal and parietal lobes [17]. Moreover, the use of mixed reality devices can increase a user's cognitive load and, in turn, increase postural sway [20, 21].

Balance is commonly assessed by measuring postural sway, or the natural and unavoidable movements that occur while standing. Postural sway is commonly assessed using a force plate or balance board to track movement by collecting and calculating either the center of pressure mean speed or an anteroposterior mean velocity [9, 12]. While center of pressure mean speed is typically used to determine instability, anteroposterior mean velocity is often used when users are at higher risk of falls, such as in studies involving the elderly [17].

3. RELATED WORKS

3.1. Industrial Augmented Reality Applications

Prior studies suggest AR is promising increased relevance to the construction industry, and as AR technology improves, its usage in the industry (and many others) is expected to increase [22-24]. The increased interest is in part due to the benefit of gesture-limited access to information while using AR to assist in tasks within maintenance and construction [5, 6]. AR can project digital step-by-step instructions directly within a worker's line of sight, which reduces the necessary head movements associated with reading traditional instructions [4]. Importantly, head movements have been linked to causing postural sway through disturbances to the vestibular systems, which are located within the inner ear [14, 25]. Many AR interface require head movements, especially due to current displays' limited field-of-view–thus understanding the impact of AR usage on postural sway can lead to greater insights on the safety of using AR at heights.

Within industrial settings, AR is also finding use in inspection, visualization, and safety training. During building and infrastructure safety inspections, AR can supplement an inspector with deep learning capabilities that allow for real-time damage detection of critical infrastructure [26]. Modern AR applications for inspection leverage convolutional neural networks to identify objects in the user's field of view, including machinery, objects, and structural defects. This framework allows for signs of corrosion and fatigue, like cracks and rusting, to be caught automatically and flagged for the inspector to investigate. This assistance can lead to greater fault detection and more thorough building and infrastructure inspections, improving public safety.

When the utilities and structures such as columns behind walls and floors are of interest, AR has been used to map out utilities in areas that cannot be viewed easily through a simulated form of "x-ray vision" [1]. Technicians and construction workers can see exactly where pipes, electrical lines, and rebar are located behind walls with this technology, preventing damaging utility lines when working around them [27]. This kind of assistance can be enhanced with a real technician being able to see exactly what another worker is seeing. Through "tele-assistance", an expert can remotely watch a technician's live camera feed and provide verbal feedback, as well as interactive virtual instructions through 3D models or text that can overlay directly to the technician's view point [19, 28].

AR has been used to improve workers safety by allowing workers to gain experience dealing with situations that would otherwise be difficult or dangerous to simulate [29]. Workers who experience these scenarios in a first-hand presentation develop a more proactive approach to preventing and managing risks [30]. In respect to ladders, augmented reality has used for ladder safety assessments by analyzing workers' postures in real time [31]. This technology is important for worker safety due to the prevalence of ladder misuse in workplace related injuries [32].

3.2. Occupational Fall Hazards and Ladder Safety

Many falls from ladders are preventable with common causes including incorrect ladder angle, improper ladder selection for the job, insufficient ladder inspection, overreaching, and lack of training [32]. We posit that without care, the use of AR on ladders could result in overreaching, since interacting with AR at far distances is difficult due physiological issues related to vergence as well as the poor ergonomics associated with interaction far from the body's center of mass. The CDC notes that novice workers (due to inexperience) and older workers (due to reduced sensory capabilities) are at higher risk of fatal falls. Such risks could be exasperated due to AR usage, especially given that AR shows promise for upskilling novice workers in new work settings [32]. Some AR applications designed for the construction site are implementing an "early warning system" that relies on computer vision to identify hazards in a worker's field of view, like floor openings or improper ladder angles, to help keep less experienced workers safe [21, 27].

3.3. Physiology on Head-Mounted Displays

When using VR head-worn displays (HWDs), it's been well documented that the mismatch between visual cues of movement from the device and the body's vestibular system not detecting movement is a leading cause of visually-induced motion sickness [14, 33]. Many HWDs block or reduce a user's peripheral vision, which is another visual cue that the body uses to maintain balance [9, 15]. Extended use of AR HWDs can result in eye strain and blurred vision, with more dynamic visuals inducing dizziness and ataxia [9, 14, 33]. Currently, some technological limitations can lead to nausea and dizziness, namely latency, lag, and frame rate modulation [14]. Issues of latency and lag are further exacerbated in AR applications, since virtual objects are expected to be accurately registered in the real world.

A significant source of eye strain can be attributed to the vergence-accommodation conflict (VAC) [14]. VAC occurs when using AR and VR HWDs because it separates the eyes' natural ability to converge on an object (convergence) with the reshaping of the lens (accommodation) [14]. When viewing an object, our eyes converge on it, and our eyes accommodate the object's distance from us. When using AR and VR, an object can appear to be within the environment at varying distances, when in reality, our focal distance is fixed to the screens close to the eyes. This has been known to cause oculomotor disturbances and is mentally taxing to our perception [14].

Research indicates that HWDs can increase postural sway when compared to viewing a traditional monitor [14, 15]. Two competing theories exist to explain this phenomenon: sensory conflict theory and postural instability theory. The long standing and more popular sensory conflict theory states that sickness and instability while using AR and VR devices is rooted in the mismatch of visual motion cues and the body's vestibular system understands that the body is not moving, much like motion-sickness that is experienced in a moving vehicle [14, 34]. Conversely, postural instability theory suggests that increases in postural sway brought about through new environmental factors have strong connections to cybersickness, which causes the symptom akin to fatigue, hangovers, and influenza, and lasting for some time after the headset is removed [34]. This theory in itself is derived from the evolutionary poison theory, which states that nausea induced from AR and VR is akin to hallucinations induced by ingesting toxic substances [34].

From a subjective viewpoint, when using HWDs users report increased fear of falling and being more prone to falling, particularly when virtual objects are not anchored in the real-world [33]. Studies have also pointed to a strong correlation between visually influenced body sway and perceived disorientation [11]. The use of a static rest frame, such as a subtle frame or crosshair, that remains fixed in a user's view provides an environment-referenced anchor that has been proven to improve a user's balance through reducing the illusion that the environment is moving and not the body [15]; however, these tools are currently not widely implemented in AR applications. These observations further motivate our project to investigate the usage of AR in scenarios of elevated fall risk.

4. METHODS

4.1. Experimental Overview

The study presented herein employs a within-subjects design consisting of a 2 (Presentation Style: world-relative, screen relative) × 3 (Task: text reading, button poking, block assembly) factorial, along with an additional block sorting task in which presentation style is not applicable (see

Section 4.4). Each in-person session will take approximately an hour and a half to complete. The order of presentation of the seven experimental conditions was determined using a balanced Latin square, which generated fourteen unique experimental condition sequences. Each of the seven experimental conditions is performed three times for a total of twenty-one trials. Tasks are performed on a ladder with participants maintaining three points of contact, uniform foot placement on a prespecified rung, and with a rest break after every seven trials. One baseline balance assessment is conducted prior to experimental trials. A force plate beneath the ladder collects center of pressure data over each 30 second trial. Participants use a Microsoft HoloLens 2 head-worn mixed reality display to assist in the six visual tasks (and kept the headset on for the seventh verbal task). Participants use standardized sneakers to avoid any effects of differing footwear.

4.2. Participants

A biological sex-balanced convenience sample of twenty-eight participants was recruited from the university and local community using flyers posted around campus and email listservs. The biological sex-balancing was achieved through asking participants of their biological sex assigned at birth and gender identity was not collected, thus the term "sex" will be used in this paper. A sample of twenty-eight participants was selected to ensure each sequence of experimental conditions was performed more than once and by resource availability. Participant body height and mass were self-reported. Participants were asked for their handedness, as well as their experience with ladders and with VR at three levels: 0-4 uses/year, 5-14 uses/year, and 15+ uses/year.

Participant inclusion criteria include: (1) having no self-reported musculoskeletal injuries within the past 6 months, (2) not heavier than 100 kilograms, (3) not taller than 2 meters, (4) do not wear glasses, though vision corrected with contact lenses was allowed, (5) do not fear heights, and (6) were not pregnant. Participants were excluded if they had recent musculoskeletal injuries or are pregnant to reduce the risk of harm to vulnerable populations. Participants were excluded for height and weight based on safety recommendations of the A-frame ladder being used. Participants were excluded if they wear glasses due to concerns of the HoloLens fitting properly. All participants were required to provide written consent prior to participation, and the study was approved by the Virginia Tech Institutional Review Board.

4.3. Experimental Set-Up

Figure 2. Ladder and force plate set-up. The force plate is beneath the metal mounting plate. The baseline balance fixation target can be seen on the wall in the top right corner.

The experimental protocol consists of two stages: a short task familiarization portion followed by experimental trials on the ladder. During the task familiarization phase, participants are first shown a presentation detailing exactly what they will see when using the HoloLens during the experiment. Once completed, the participant calibrates the HoloLens's eye tracking from instructions delivered by the experimenter. They then stand on marked vertical lines on the ground to simulate proper foot placement while on the ladder, while standing in front of the assembly box. Participants are walked through a practice sequence, performing each of the three tasks once while on the ground to become familiar with the tasks themselves. The results of the practice trials will not be analyzed.

Experimental trials are then performed on a ladder on top of a force plate. Two climbing mats are placed next to the ladder in case of a lateral fall. The ladder is secured to metal plate using four bolts at each foot, and the metal plate is bolted directly to the force plate. Participants climb the ladder up to a designated rung to complete the trials.

4.4. Protocol

Before any AR use, participants complete a baseline balance trial on the ladder involved holding the top of the ladder with only their dominant hand with their nondominant hand at their side. Participants are instructed to stand as still as possible, without talking, while staring at a fixation target recommended for stability testing [35]. Each baseline collected 30 seconds of balance data at 500 scans/second. Once completed, participants are instructed to descend the ladder.

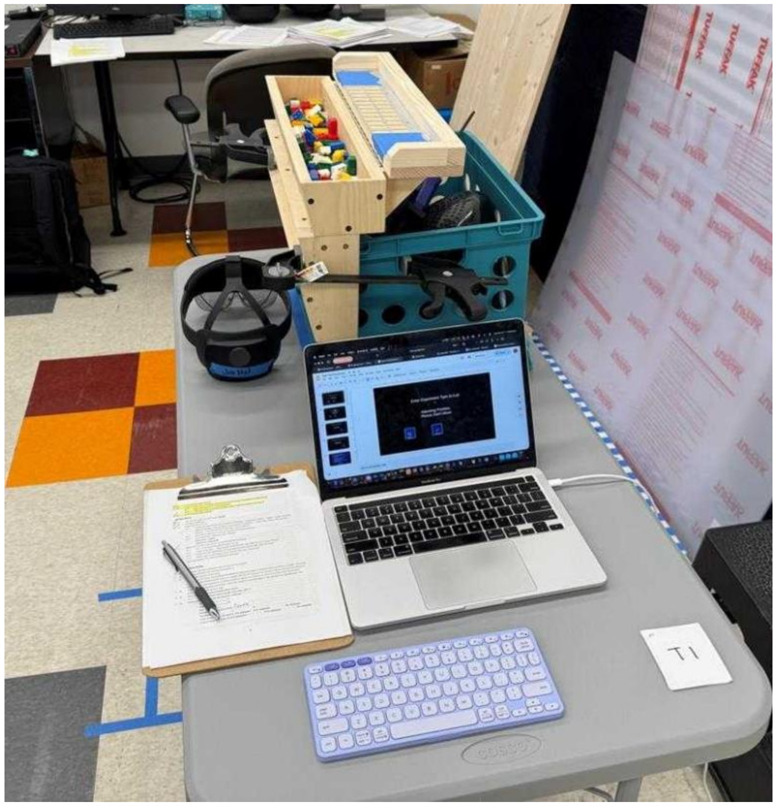

Figure 3. Task familiarization set-up from experimenter's point of view. The laptop is used to present a short presentation to the participants showing them exactly the screens they will see during the experiment.

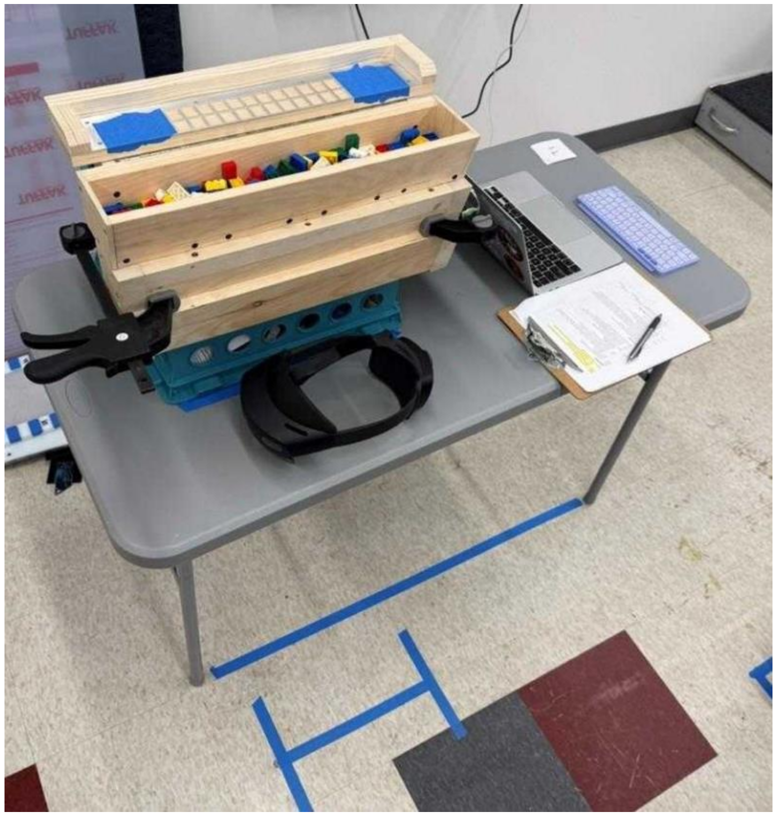

Figure 4. Task familiarization set-up from participant's point of view. The "H" taped on the floor represents where the participants feet are to be places, with the horizontal line representing the width of the ladder rung.

Participants are asked to don and adjust the HoloLens to ensure a proper fit. Participants are then instructed to look at their hands through the device to ensure their hands are being properly tracked before the experiment begins. The experimenter walks participants through the eye calibration process and has them raise the visor after completion. At this point, a short presentation detailing the exact screens that are displayed in the experiment are shown to participants, so they knew exactly what to expect when the experiment begins and how to respond to certain visual prompts. The experimenter will navigate the HoloLens app from a Bluetooth-enabled keyboard, as we did not want to rely on the HoloLens's built-in voice command recognition. After the familiarization view, participants walk around to a taped "H" on the ground, position their feet in the same way they will need to later the ladder, and begin the interactive phase of task familiarization. This phase begins with a brief calibration (that establishes the world relative frame of reference for each participant based on head position) and then presents each task and presentation style once so participants understand what actions are needed to successfully complete each trial. Once this phase is complete, participants raised the visor and move on to the formal experimental trail phase.

To begin the experimental trials, researchers first remove the assembly box (Figure 4) from the task familiarization table and clamp it to the top of the ladder. The force plate beneath the ladder is then calibrated with the box attached and zeroed out, after which each participant climbs the ladder to the rung marked with tape. Participants then lower the visor, and the AR testbed application is launched remotely from the experimenter's laptop. Participants then perform a second calibration to relocate the world-relative frame of reference based on the participants' new head position atop the ladder. Once the experimenter verbally confirms that the participant is ready, the AR testbed application begins the task sequence.

As previously mentioned, participants complete seven experimental conditions: a reading, block assembly and button poking task presented separately as either screen-relative or world-relative graphics, as well as an additional block assembly task where the experimenter conveys assembly order verbally. Each task and input condition are listed in Table 1.

Table 1. Seven experimental conditions to be completed on the ladder.

Tasks	**Input**
Text Reading	Visual via AR, screen-relative
Text Reading	Visual via AR, world-relative
Poking Virtual Buttons	Visual via AR, screen-relative
Poking Virtual Buttons	Visual via AR, world-relative
Assembling Physical Blocks	Visual via AR, screen-relative
Assembling Physical Blocks	Visual via AR, world-relative
Assembling Physical Blocks	Verbal input from experimenter

These seven experimental conditions were chosen to simulate diverse tasks that could be completed on a ladder with differing degrees of user demand. The task that requires the least integration with the AR was text reading, meant to simulate reading instructions or a manual. A task with increasing user demand is the button poking task, meant to simulate AR menu navigation. The task with the most user demand is the block sorting task, as it required both referring to displayed information and performing a fine-motor task. The block assembly with verbal input simulates a partner instructed task and was designed to see if a block assembly task with verbal input would be more stable than with AR input.

All AR interfaces are presented ~55cm from the user's head with the world-relative presentation style anchoring based on the calibration phase. Once anchored, the world-relative interfaces do not move. This distance was determined during pilot testing and aimed to balance participants' ability to reach the button poking interface comfortably while also presenting other task interfaces at comfortable distances for both reading and assisting in the assembly task.

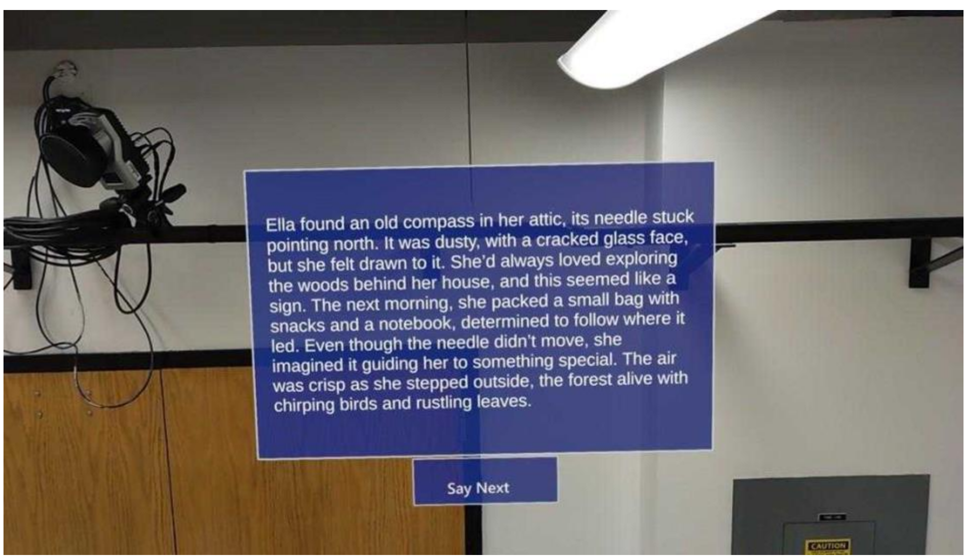

Figure 5. Participant's point of view of the text reading task.

Text Reading Task. During the reading tasks, participants are instructed to speak clearly and loudly, as if they are giving a speech or a presentation. A paragraph is displayed via HoloLens which participants read aloud. Three paragraphs containing four to six sentences per trial were generated with OpenAI's ChatGPT using a prompt for a warm, fantasy story in English. Once they reach the end of each passage, participants are instructed to say "Next" to cue the experimenter to manually administer the next prompt.

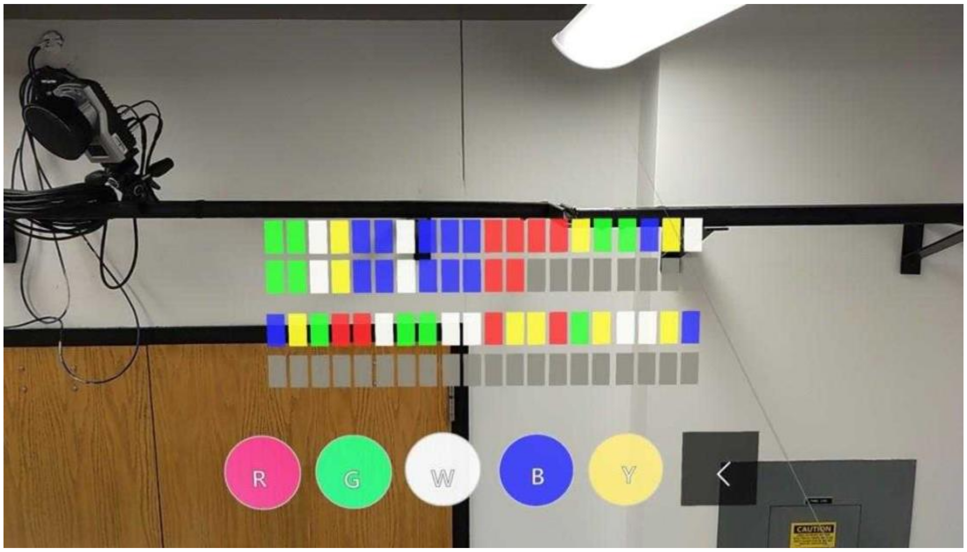

Figure 6. Participant's point of view of the button poking task.

Button Poking Task. During the interactive button poking task, participants use their dominant hand to match a color sequence of rectangles as presented via AR by poking corresponding circular virtual buttons located beneath the sequence. The sequences include red, yellow, green, blue, and white squares, and the buttons available reflect these five colors. Participants are also given a backspace button and told that use of the backspace is encouraged in the event of a mistake. As with the previous tasks, if participants complete the task before the forty second period was complete, they say "Finish."

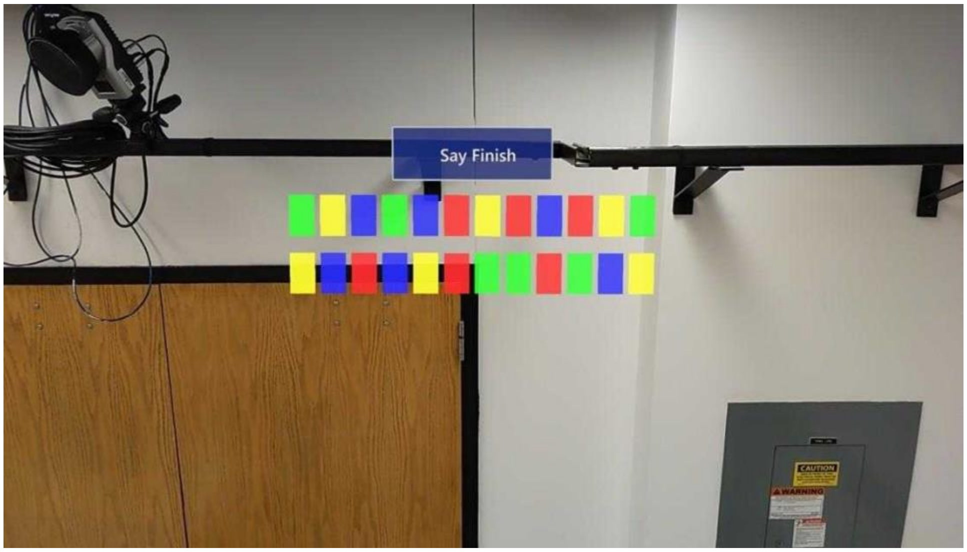

Figure 7. Participant's point of view of the block assembly task.

Block Assembly Task. During the block assembly tasks, participants use their dominant hand to place the blocks in the sorting platform in the exact same order shown via the AR display. The block sequences displayed includes red, yellow, green, and blue blocks in randomized orders. The physical blocks available to participants include the same four colors, as well as white blocks to act as a hinderance. In the case of the participant completing the task before the forty second period is complete, they are instructed to say "Finish." The sorting platform is then dumped back into the block bin in preparation for future trials.

The block sorting task that uses verbal instructions follows the same protocol as above, but instead of the participant using AR visual information, the experimenter reads a color sequence aloud – reading one color at a time as participants assembled blocks. The next color is spoken when the experimenter sees the previous block placed in a location. In this condition, participants wore the HoloLens display, but there is no visual information required for the task nor displayed via the display.

Upon completion of trial twenty-one, participants descend the ladder, return the shoes, and are compensated for their time.

4.5. Dependent Measure

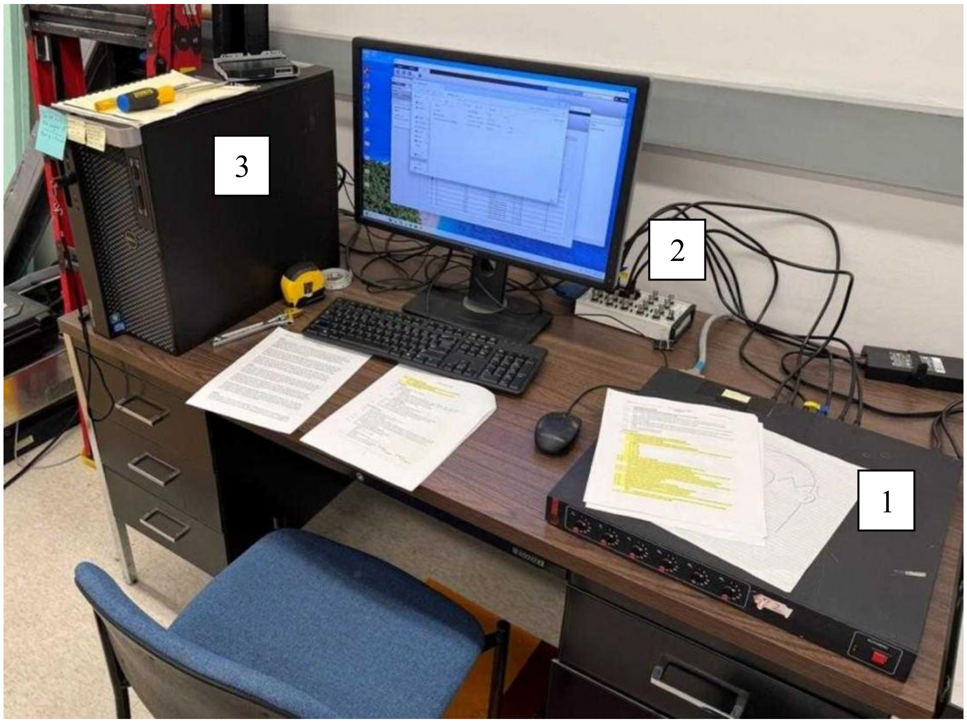

Figure 8. Data collection computer set-up. The amplifier (1) sends voltage signals to the data acquisition system (2) which sends the signals to the desktop computer (3).

One dependent variable will be used to evaluate the effect of different tasks—postural sway. Postural sway is determined using the mean speed of the center of pressure measured by the force plate beneath the ladder. The center of pressure (COP) is the location on the surface of the force plate that represents the mean position of forces applied. As a participant stands and sways on the ladder, the COP on the force plate moves with them. The mean speed of the COP during trials is a common and sensitive measurement of postural sway and fall risk [12, 16, 36]. The center of pressure mean speed data is collected with a force plate [Model 4550-08, Bertec Corp. Columbus, OH, USA] beneath the ladder set-up. The voltages collected from the force plate are amplified [Model AM-6100, Bertec Corp. Columbus, OH, USA] and sampled at 500 Hz using a data acquisition system [National Instruments, Austin, TX, USA] for post-processing. COP mean speed is calculated using customized MatLab™ code. To isolate the effects of the tasks on postural sway and to account for individual differences, the mean COP speed from each participant's baseline balance trial will be subtracted from the mean COP speed of each experimental trial.

4.6. Analysis

One dependent variable was calculated to evaluate the effect of different tasks on a ladder on postural sway—COP mean speed. This value will be calculated through a MatLab™ script that uses the COP at each data point and compares them against each other to determine mean speed. Statistical analysis will be conducted using *JMP*. A mixed-effects model with REML estimation will be used to analyze the effects of presentation style, task, and their interaction on participants' increase in postural sway from their baselines.

The statistical analysis addressed the hypotheses using a two-way ANOVA with factors of task, presentation style, task × presentation style interaction, sex, sex × task, sex × presentation style, repetition, experience with VR, and experience climbing ladders. The dependent variable for this analysis with the difference in COP mean speed from baseline. *A priori* pairwise comparisons were performed using Tukey's HSD.

5. RESULTS

Four participants were excluded from analysis due to equipment malfunction or participants misinterpreting instructions. Statistical analysis was conducted using JMP Statistical Discovery. A mixed-effects model with REML estimation was used to analyze the effects of presentation style, task, and their interaction on participants' increase in postural sway from their baselines. Data that did not pass tests for normality was transformed with a square root transform.

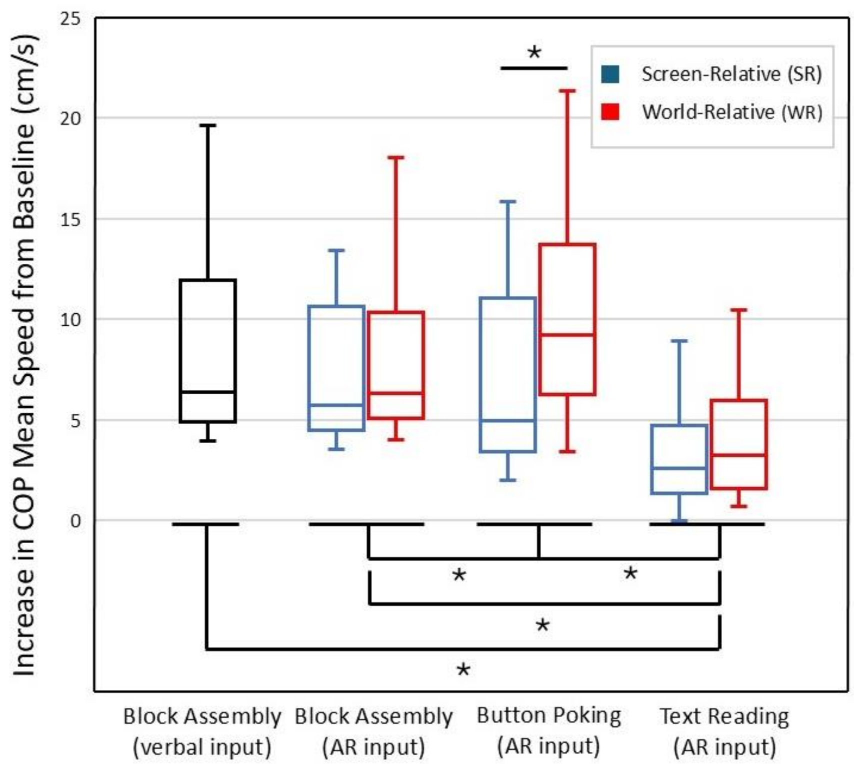

Figure 9. Increase in postural sway from baseline across tasks and presentation styles. Brackets above the box and whisker plots indicate differences between presentation styles within each task, and brackets below the box and whiskers plots indicate differences between tasks (inclusive of both presentation styles). All asterisks over brackets indicate statistical significance at $p < 0.05$. Error bars show 95% confidence intervals.

Figure 9 uses box and whisker plots to illustrate the median, upper, and lower quartiles, and range of each of four tasks.

5.1. Main Effects of Task

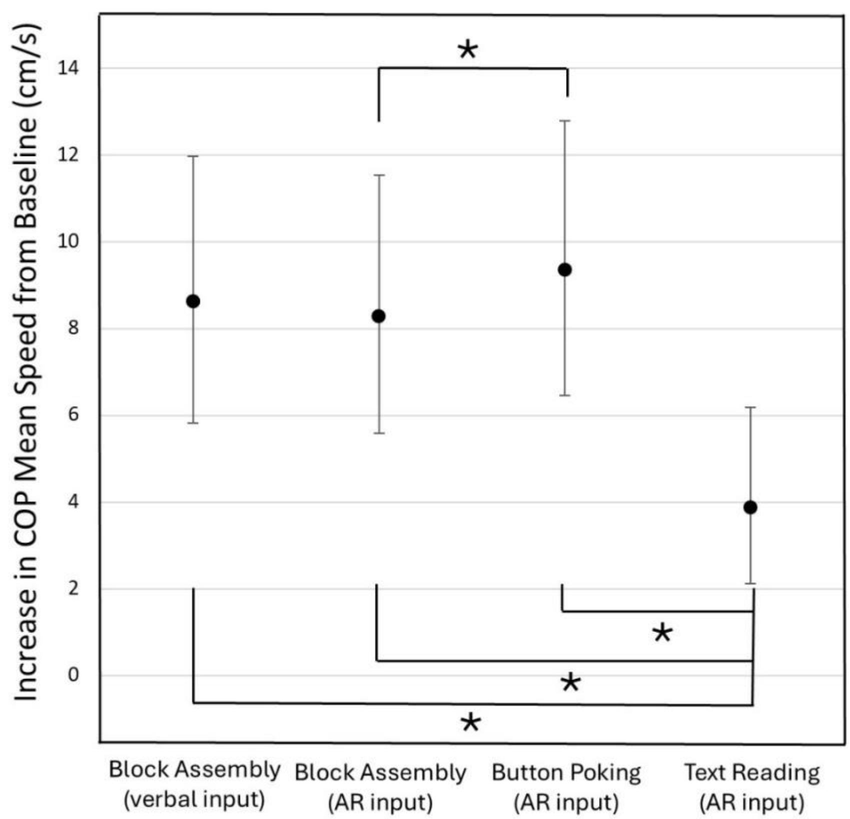

Figure 10. Increase in postural sway from baseline across tasks. Tasks with multiple presentation styles were combined for analysis against block assembly with verbal input. All asterisks over brackets indicate statistical significance at $p < 0.05$. Error bars show 95% confidence intervals.

A main effect of task was observed, $F(2,398)=172.49$, $p<0.0001$. Pairwise comparisons using Tukey HSD indicated the task being performed has a measurable effect on a participant's postural sway when compared to each participant's baseline balance measurement. To conduct this comparison with the block assembly with verbal input, world-relative and screen-relative presentation styles were combined. Text reading tasks showed statistically significant differences between all other tasks, and button poking and block assembly with AR input displayed a statistically significant difference.

5.2. Main Effects of Presentation Style

There was a main effect of presentation style on COP mean speed, $F(1,398)=58.02$, $p<0.0001$. Pairwise comparisons using Tukey HSD indicated the world-relative presentation style increased postural sway (an adverse effect on balance) more than the screen-relative presentation style when compared to each participant's baseline balance measurement. Block assembly with verbal input was excluded from this analysis due to not having either world-relative or screen-relative presentation styles.

5.3. Interaction Effects

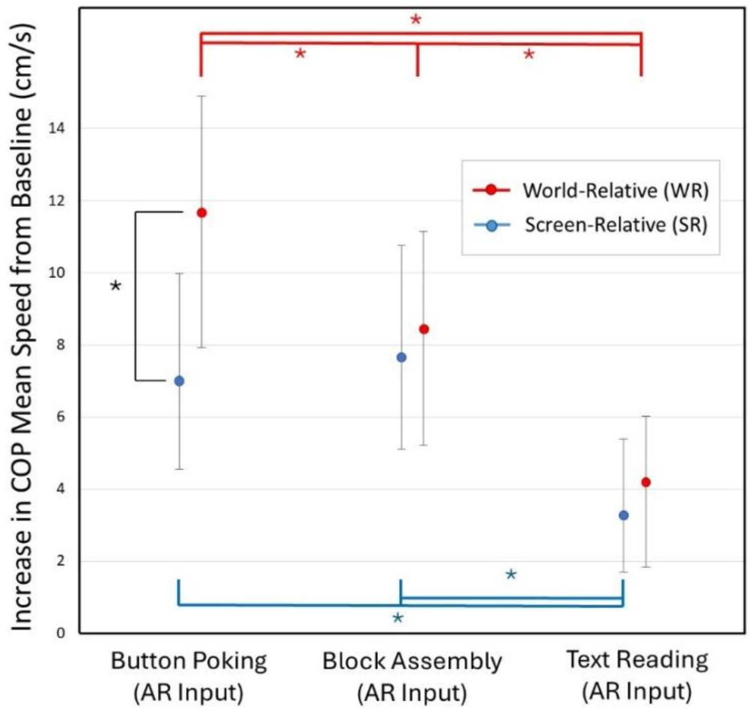

Figure 11. Increase in postural sway from baseline across tasks and presentation styles. Brackets are color-coded for each presentation style.

The main effects are superseded by a significant Task × Presentation Style interaction $F(20,398)=14.28$, $p<0.0001$ indicating the effect of presentation style on COP mean speed differed across tasks. Across both world-relative and screen relative presentation styles, the button poking task with AR input demonstrated the greatest postural sway deviation from the baseline when compared to the text reading with AR input and both forms of the assembly tasks. In particular, the button poking task presented in a world-relative presentation style showed significantly higher sway than all other task and presentation style combinations. Table 2 illustrates the differences in COP mean speeds, and the lower and upper confidence limits of the difference. Block assembly with verbal input was excluded from this analysis due to not having a world-relative or screen-relative presentation style.

Table 2. Least squares means differences between world-relative button poking and all other task and presentation style combinations.

Task/Presentation Style	Δ mean	Δ Lower CL	Δ Upper CL
Button Poking (AR)/WR	**-**	**-**	**-**
Button Poking (AR)/SR	**0.776**	**0.521**	**1.030**
Block Assembly (AR)/WR	**0.488**	**0.233**	**0.743**
Block Assembly (AR)/SR	**0.634**	**0.380**	**0.888**
Text Reading (AR)/WR	**1.350**	**1.096**	**1.605**
Text Reading (AR)/SR	**1.605**	**1.349**	**1.861**

Text reading tasks had the lowest postural sway increase from baseline overall; both screen-relative and world-relative presentation styles differed significantly from all button poking and block assembly tasks. The two text reading tasks were also not statistically significant from one another (Δ mean = 0.255, Δ std err = 0.090, 95% CI [-0.002, 0.511]), suggesting presentation style had negligible effect on postural sway when reading text. In contrast, the button poking task in a world-relative presentation style showed significantly greater postural sway increases from baseline than in the screen-relative presentation style, suggesting that the world-relative presentation style was disorienting when participants were engaging with buttons in the AR space. The assembly task with AR input also presented this trend of the world-relative presentation style being less stable than the screen-relative presentation style, though the difference between presentation styles was not statistically significant.

5.4. Other Findings

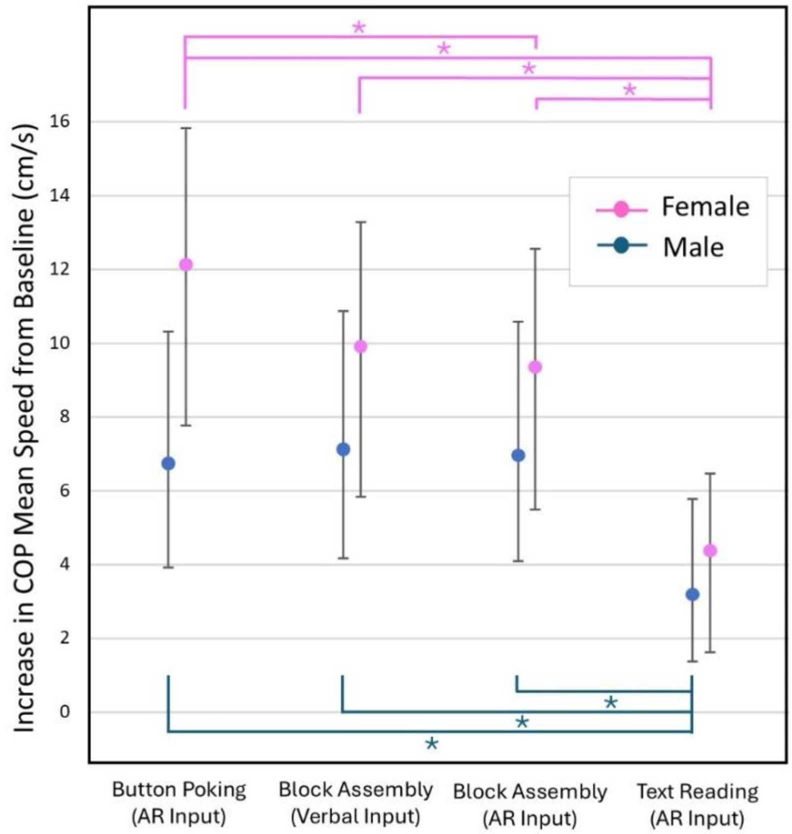

Figure 12. Increase in postural sway from baseline across tasks between male and female participants. Brackets are color-coded to differentiate males and females.

Secondary analyses included sex as an independent variable, which revealed significant Sex × Task ($F(2,413)=13.91$, $p<0.0001$) and Sex × Presentation Style ($F(1,413)=5.78$, $p=0.017$) interactions despite no significant main effect of sex on COP mean speed ($p = 0.097$), meaning that detectable difference in COP is due to differences in COP across tasks and presentation styles, not sex.

When comparing performance across tasks and sex, we observed significant differences in COP mean speed between text reading (AR input) and block assembly (both verbal and AR input), as well as text reading (AR input) and button poking (AR input). However, there was a significant difference between block assembly (AR input) and button poking (AR input) for female participants that was not seen in male participants.

With regards to presentation style, trends suggest female participants saw a larger increase in postural sway from their baseline under the world-relative presentation style than the screen-relative presentation style (LS Means difference = 0.50) when compared to male participants (LS Means difference = 0.26). This suggests that presentations style may affect postural stability differently dependent on sex, though a thorough analysis of these effects is outside the scope of this study.

The fixed effect tests showed that the two measures of prior experience, experience using ladders ($F(1,21)=0.63$, $p=0.438$) and experience using VR ($F(2,21,)=0.88$, $p=0.431$) did not have an effect on postural sway. Trial number also did not have an effect on postural sway ($F(20,398.2)=1.48$ $p=0.086$), suggesting that no systematic order effects due to, for example, learning or fatigue were seen across trials.

6. DISCUSSION

The purpose of this study was to investigate the effects of AR on balance during simulated construction tasks while on a ladder. We explore how we accomplished this in reference to our research questions and hypotheses.

6.1. Research Questions

RQ1: What is the effect of simulated AR work tasks on postural sway while working on a ladder? To address this question, we look back at the data we collected and the main effects of task on postural sway. Task had the strongest effect in the entire model ($F(2, 398)=172.49$, $p<0.0001$), suggesting that the physical and/or AR-based tasks that a worker is doing while on a ladder play a key role in their postural stability. The button poking tasks saw the greatest increase in postural sway from baseline, suggesting that interacting with AR interfaces (that require virtual button presses) may cause instability, and at least is associated with more instability than reading AR text (and more instability than the block assembly task for world-relative condition). The block assembly task saw moderate postural sway increase, and the text reading task saw the least postural sway increase. Providing visual instruction visually via AR did not result in any measurable differences in postural sway as compared to providing oral instructions.

To understand this behavior, two explanations are considered: reaching and cognitive load. Leaning while using a ladder is a known safety concern for falls, and our button poking task required the greatest anterior reaching of the three tasks. The literature suggests that participants are more likely to physically maneuver their body's to a posture that allows for the easiest interactions with the AR interface [32, 37, 38], thus we expected that users were more likely to lean forward to poke the AR interface, and when doing so, may introduce more mediolateral sway (side-to-side) due to the fact that A-frame ladders are prone to more lateral instability than longitudinal (front-to-back). While the block assembly required more hand movements, and constant checks to visual instructions, the hand movements did not require reaching and were limited to the assembly box space located directly atop the ladder. Note that this location is closer to participants' (and ladder's) center of gravity. If the assembly task had been placed in front as opposed to atop of the ladder (for example, at the same distance as the poking task), different results may have been observed.

Further, we believe that the use of the assembly box may have given a visual and physical anchor fixed to the ladder for participants to retain balance, whereas the AR interface used for the poking task had no similar affordances. Research indicates that cognition is a component in postural control, and when mental facilities are engaged, postural sway increases with less bandwidth to be applied to proprioceptive responses [39]. The "posture first" principle dictates that individuals will prioritize balance over a cognitive task, though this can be transgressed if a participant has a sense of safety, such as maintaining three points of contact or being in a laboratory setting [39, 40].

The results we collected and the justifications we have found imply that tasks that involve interacting with an AR interface can lead to greater instability as compared to simply standing on a ladder. This should be considered in the development of AR applications for use in the construction industry, as more interaction with an AR interface a worker does while on a ladder can lead to greater postural sway and thus risk of imbalance.

RQ2: Do the effects of simulated AR work tasks on postural sway differ between screen-relative and world-relative presentation styles? From our data, we saw a highly significant effect of presentation style suggesting that how information is presented in space can have a meaningful influence on a worker's postural sway while on a ladder. Most of these effects are attributable to button poking task where the world-relative presentation style produced higher postural sway than screen-relative, implying that when an interaction heavy AR interface stays fixed within the environment, participants displayed greater postural sway.

When analyzing these results, we explored the motion participants exhibited on the ladder. When tasks were presented in a screen-relative presentation style, users kept their heads very stable to clearly see the AR projection. In order to focus on colors and text, the eyes rely on foveal vision, located directly in the center of our field of vision [41, 42]. When the head must be kept stable, we believe viewing the AR input requires the user's eyes to scan without moving their head. In a world-relative presentation style, the user is free to move their head while parsing visual information and can even move their head and shoulders enough to where the motions required to poke the interface become more exaggerated than those associated with screen-fixed buttons. For example, if users stand tall and do not lean into the ladder while using the world-relative condition they will need to reach further in front of themselves. Further, studies have demonstrated that when information is presented in a format wider than our direct parafoveal vision, head movement is significantly increased [43]. This kind of head movement can lead to greater postural sway from perturbations to the vestibular systems [9, 17].

While the difference between postural sway in world-relative and screen-relative presentation styles were similar for the block assembly and text reading tasks, they varied significantly in the button poking task, as seen by the significant interaction effect. This interaction effect implies that care must be taken in the design of AR applications for use in the construction industry where ladder is common. Applications requiring poking should consider screen-relative displays that do not obstruct a worker's vision but maintain their position relative to the worker's head orientation. World-relative presentation styles should be reserved for AR interactions on ladders that do not require reaching, or for construction tasks that not performed on ladders [23, 24].

6.2. Hypotheses

Hypothesis 1: Compared to other AR tasks, the increase in postural sway from baseline will be the largest during the interactive button-poking task. This hypothesis was supported by observation of the main effect of task from the mixed-effects model. Pairwise comparisons confirm that the button poking task demonstrated the greatest average postural sway deviation from baseline (9.35 cm/s) when compared to block assembly with AR input (8.30 cm/s), block assembly with verbal input (8.63 cm/s), and text reading (3.88 cm/s). To rationalize this finding, the explanation suggested with research question 1 suggests a combination of participants being forced to lean forward to interact with the interface, as well as an increase in cognitive load from interacting with a fully virtual interface.

Hypothesis 2: The increase in postural sway from baseline during AR tasks will be larger with a screen-relative presentation style than with a world-relative presentations style. This hypothesis was refuted by main effect of presentation style from the mixed-effects model, since tasks with the world-relative presentation style demonstrated greater postural sway than the tasks with the screen-relative condition across all AR tasks. While the difference was not statistically significant for the text reading and block assembly tasks, the button poking task displayed a statistically significant difference between world-relative and screen-relative presentations styles.

As explored with research question 2, we believe that participants held their heads more steady in order to accurately read from and input commands into the screen-relative tasks, as opposed to the world-relative tasks, in which head position and movement are irrelevant.

6.3. Other Findings

Considering the findings of sex × task and sex × presentation style interaction effects, these patterns suggested that any differences in COP mean speed between males and females was due to COP mean speed differences between tasks and presentation styles. Nonetheless, the trends observed between sexes may have contributed to the interaction effect between task × presentation style. Further research into the differing effects of tasks and presentations styles on postural sway between sexes may reveal trends not seen in this study.

6.4. Limitations and Future Work

Limitations of this study include the sample demographics not aligning with demographics of construction workers who would be using AR technology in the field. The demographics of workers at a higher risk of ladder falls includes older Hispanic males and self-employed workers [44]. A wider range of participant ages would generate more generalizable results. The study was conducted in a laboratory setting, and therefore does not include typical workplace environmental factors, such as noise, temperature, humidity, and other distractors. The ladder assembly, while useful to simulate an assembly task at the top of a ladder, does not simulate a work task such as installation or wiring that involves working against a surface next to the ladder or overhead. The study also does not collect subjective balance data, such as perceived instability or symptoms of cybersickness, which could provide useful insights into confidence and perceptions of safety.

While all tasks were presented directly in front of the participant, this obscures the user’s view. A follow-up involving world-relative presentation could place the display at various locations around participants field of view, encouraging them to rotate their head greater angles to view information. This study focused on interacting with AR interfaces using gesture controls like poking. A further exploratory study could explore the difference in postural sway between a task completed with gesture controls and voice controls. The implications of such further research can help advise AR application design for worker safety.

7. CONCLUSION

While augmented reality is being developed as tool for the construction industry, its effects on postural sway while using a ladder is understudied. In this study we designed four tasks to be completed while on a ladder within two presentation styles and measured the change in participant’s postural sway across tasks and presentation styles from a baseline. We used an A-frame ladder on top of a force plate to measure center of pressure mean speed, an indicator of postural sway. We found significant main effects of presentation style and of task on postural sway increase from baseline balance as well as an interaction effect. Increase in postural sway from baseline balance was greatest during a button poking task in a world-relative presentation style. Our results aim to shed light on the potential risks of AR usage while on a ladder, as well as guide AR application development for the construction industry with safety standards in mind.

8. ACKNOWLEDGEMENTS

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9. REFERENCES

[1] R. T. Azuma, "A survey of augmented reality," *Presence: teleoperators & virtual environments,* vol. 6, no. 4, pp. 355–385, 1997.

[2] D. Van Krevelen and R. Poelman, "A survey of augmented reality technologies, applications and limitations," *International journal of virtual reality,* vol. 9, no. 2, pp. 1–20, 2010.

[3] A. Heinzel, S. Azhar, and A. Nadeem, "Uses of augmented reality technology during construction phase," in *Conference: 9th International Conference of Construction in the Twenty-first Century (CITC9), Dubai, United Arab Emirates*, 2017, pp. 1–8.

[4] S. Henderson and S. Feiner, "Exploring the benefits of augmented reality documentation for maintenance and repair," *IEEE transactions on visualization and computer graphics,* vol. 17, no. 10, pp. 1355–1368, 2010.

[5] A. Malta, T. Farinha, and M. Mendes, "Augmented reality in maintenance—history and perspectives," *Journal of Imaging,* vol. 9, no. 7, p. 142, 2023.

[6] A. Seeliger, L. Cheng, and T. Netland, "Augmented reality for industrial quality inspection: An experiment assessing task performance and human factors," *Computers in Industry,* vol. 151, p. 103985, 2023.

[7] B. o. L. Statistics. "Fatal injuries from ladders down in 2020; nonfatal ladder injuries were essentially unchanged." U.S. Department of Labor. https://www.bls.gov/opub/ted/2022/fatal-injuries-from-ladders-down-in-2020-nonfatal-ladder-injuries-were-essentially-unchanged.htm (accessed March 4, 2026).

[8] H. J. Lipscomb, A. L. Schoenfisch, and K. S. Shishlov, "Non-fatal contact injuries among workers in the construction industry treated in US emergency departments, 1998-2005," *Journal of safety research,* vol. 41, no. 3, pp. 191–195, 2010.

[9] P. Epure *et al.*, "Effect of the Oculus Rift head mounted display on postural stability," in *The 10th International Conference on Disability Virtual Reality & Associated Technologies: Proceedings*, 2014: Reading University Press, pp. 119–127.

[10] C. L. Hughes, C. Fidopiastis, K. M. Stanney, P. S. Bailey, and E. Ruiz, "The psychometrics of cybersickness in augmented reality," *Frontiers in Virtual Reality,* vol. 1, p. 602954, 2020.

[11] B. D. Lawson and K. M. Stanney, "Editorial: Cybersickness in Virtual Reality and Augmented Reality," (in English), *Frontiers in Virtual Reality,* Editorial vol. Volume 2 - 2021, 2021–October–13 2021, doi: 10.3389/frvir.2021.759682.

[12] E. Chiarovano, C. De Waele, H. G. MacDougall, S. J. Rogers, A. M. Burgess, and I. S. Curthoys, "Maintaining balance when looking at a virtual reality three-dimensional display of a field of moving dots or at a virtual reality scene," *Frontiers in neurology,* vol. 6, p. 164, 2015.

[13] S. Oviatt, "User-centered modeling and evaluation of multimodal interfaces," *Proceedings of the IEEE,* vol. 91, no. 9, pp. 1457–1468, 2003.

[14] V. M. Schroeder and L. A. T. Biggs, "Fatigue and Eye Strain Associated with Virtual Reality and Augmented Reality Systems," *Adapting Virtual Reality and Augmented Reality Systems for Naval Aviation Training,* vol. 4, p. 104, 2018.

[15] S. M. Shahnewaz Ferdous, T. I. Chowdhury, I. M. Arafat, and J. Quarles, "Static rest frame to improve postural stability in virtual and augmented reality," *Frontiers in Virtual Reality,* vol. 1, p. 582169, 2021.

[16] F. Quijoux *et al.*, "Center of pressure displacement characteristics differentiate fall risk in older people: A systematic review with meta-analysis," *Ageing Research Reviews,* vol. 62, p. 101117, 2020.
[17] C.-J. Chang, T.-F. Yang, S.-W. Yang, and J.-S. Chern, "Cortical modulation of motor control biofeedback among the elderly with high fall risk during a posture perturbation task with augmented reality," *Frontiers in aging neuroscience,* vol. 8, p. 80, 2016.
[18] G. Yenduri *et al.*, "Spatial Computing: concept, applications, challenges and future directions," *arXiv preprint arXiv:2402.07912,* 2024.
[19] J. M. Davila Delgado, L. Oyedele, T. Beach, and P. Demian, "Augmented and virtual reality in construction: Drivers and limitations for industry adoption," *Journal of construction engineering and management,* vol. 146, no. 7, p. 04020079, 2020.
[20] S. Blomqvist, S. Seipel, and M. Engström, "Using augmented reality technology for balance training in the older adults: a feasibility pilot study," *BMC geriatrics,* vol. 21, no. 1, p. 144, 2021.
[21] A. Z. Kolaei, E. Hedayati, M. Khanzadi, and G. G. Amiri, "Challenges and opportunities of augmented reality during the construction phase," *Automation in Construction,* vol. 143, p. 104586, 2022.
[22] S. Rankohi and L. Waugh, "Review and analysis of augmented reality literature for construction industry," *Visualization in Engineering,* vol. 1, no. 1, p. 9, 2013.
[23] S. Ahmed, "A review on using opportunities of augmented reality and virtual reality in construction project management," *Organization, technology & management in construction: an international journal,* vol. 10, no. 1, pp. 1839–1852, 2018.
[24] H. Nassereddine, A. S. Hanna, D. Veeramani, and W. Lotfallah, "Augmented reality in the construction industry: use-cases, benefits, obstacles, and future trends," *Frontiers in Built Environment,* vol. 8, p. 730094, 2022.
[25] J. Allum, M. Gresty, E. Keshner, and C. Shupert, "The control of head movements during human balance corrections," *Journal of Vestibular Research,* vol. 7, no. 2-3, pp. 189–218, 1997.
[26] S. Wang, S. A. Zargar, and F.-G. Yuan, "Augmented reality for enhanced visual inspection through knowledge-based deep learning," *Structural Health Monitoring,* vol. 20, no. 1, pp. 426–442, 2021.
[27] J. Xu and F. Moreu, "A review of augmented reality applications in civil infrastructure during the 4th industrial revolution," *Frontiers in Built Environment,* vol. 7, p. 640732, 2021.
[28] F. De Pace, F. Manuri, and A. Sanna, "Augmented reality in industry 4.0," *Am. J. Comput. Sci. Inf. Technol,* vol. 6, no. 1, p. 17, 2018.
[29] R. E. Pereira, M. Gheisari, and B. Esmaeili, "Using panoramic augmented reality to develop a virtual safety training environment," in *Construction research congress 2018*, 2018, pp. 29–39.
[30] S. Bhoir and B. Esmaeili, "State-of-the-art review of virtual reality environment applications in construction safety," *AEI 2015,* pp. 457–468, 2015.
[31] H. Iyer, J. Isingizwe, R. Eiris, and H. Jeong, "Ladder safety assessment using head-mounted 360-degree camera-based posture estimation overlayed real-time in augmented reality," in *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 2024: IEEE, pp. 1–4.

[32] N. I. f. O. S. a. Health. "Ladder Safety." Centers for Disease Control and Prevention. https://www.cdc.gov/niosh/falls/ladder/index.html (accessed.
[33] M. Kaufeld, M. Mundt, S. Forst, and H. Hecht, "Optical see-through augmented reality can induce severe motion sickness," *Displays,* vol. 74, p. 102283, 2022.
[34] S. Davis, K. Nesbitt, and E. Nalivaiko, "A Systematic Review of Cybersickness," presented at the Proceedings of the 2014 Conference on Interactive Entertainment, Newcastle, NSW, Australia, 2014. [Online]. Available: https://doi.org/10.1145/2677758.2677780.
[35] L. Thaler, A. C. Schütz, M. A. Goodale, and K. R. Gegenfurtner, "What is the best fixation target? The effect of target shape on stability of fixational eye movements," *Vision Research,* vol. 76, pp. 31–42, 2013/01/14/ 2013, doi: https://doi.org/10.1016/j.visres.2012.10.012.
[36] F. Quijoux *et al.*, "A review of center of pressure (COP) variables to quantify standing balance in elderly people: Algorithms and open-access code," *Physiological reports,* vol. 9, no. 22, p. e15067, 2021.
[37] C. L. Deschler, E. M. Pliner, D. L. Sturnieks, S. R. Lord, and K. E. Beschorner, "Correlations between reach, lean and ladder tipping risk," *Journal of biomechanics,* vol. 150, p. 111508, 2023.
[38] R. A. Partridge, A. S. Virk, and R. E. Antosia, "Causes and patterns of injury from ladder falls," *Academic emergency medicine,* vol. 5, no. 1, pp. 31–34, 1998. [Online]. Available: https://onlinelibrary.wiley.com/doi/pdfdirect/10.1111/j.1553-2712.1998.tb02571.x?download=true.
[39] J. Barra, A. Bray, V. Sahni, J. F. Golding, and M. A. Gresty, "Increasing cognitive load with increasing balance challenge: recipe for catastrophe," *Experimental brain research,* vol. 174, no. 4, pp. 734–745, 2006.
[40] G. Andersson, J. Hagman, R. Talianzadeh, A. Svedberg, and H. C. Larsen, "Effect of cognitive load on postural control," *Brain research bulletin,* vol. 58, no. 1, pp. 135–139, 2002.
[41] E. E. Stewart, M. Valsecchi, and A. C. Schütz, "A review of interactions between peripheral and foveal vision," *Journal of vision,* vol. 20, no. 12, pp. 2–2, 2020.
[42] M. Lavidor and V. Walsh, "The nature of foveal representation," *Nature Reviews Neuroscience,* vol. 5, no. 9, pp. 729–735, 2004.
[43] F. A. Proudlock, H. Shekhar, and I. Gottlob, "Coordination of eye and head movements during reading," *Investigative ophthalmology & visual science,* vol. 44, no. 7, pp. 2991–2998, 2003.
[44] C. K. Chaumont Menéndez, J. W. Collins, P. Simeonov, and C. M. Socias-Morales, "Occupational Ladder Fall Injuries–United States, 2011," 2014.